Energy-Aware Online Positioning of UAV Base Stations in Dynamic Networks

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by

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Chapter 1

Introduction

1.1 Background

Wireless cellular networks are designed to provide reliable coverage and sufficient traffic handling capacity under normal operating conditions. However, during emergency situations such as natural disasters, infrastructure failures, or large-scale public events, parts of the network may become overloaded or completely non-operational, leading to coverage gaps and degraded quality of service for users.

To address these challenges, UAV-mounted base stations (UAV-BSs) have emerged as a flexible and rapidly deployable solution. A UAV-BS can dynamically provide temporary wireless coverage in affected areas and assist overloaded regions experiencing sudden traffic demand. Its mobility allows it to reposition according to evolving conditions, making it well suited for emergency communication in dynamic wireless networks. The effectiveness of UAV-assisted communication depends critically on the positioning of the UAV-BS. A well-positioned UAV-BS can significantly improve network performance by extending coverage, reducing interference, and supporting users that would otherwise remain unserved; poor positioning may instead cause inefficient coverage and increased interference with operational terrestrial base stations. Because of factors such as unknown user distribution, changing traffic hotspots, varying propagation conditions, and the trade-off between coverage restoration and capacity enhancement, several recent works have proposed methods to address this positioning problem [1, 4, 5].

1.2 Motivation

Online UAV-BS positioning approaches have demonstrated significant improvements in network performance by dynamically adapting the drone location according to evolving wireless conditions, improving coverage and maintaining service quality during network disruptions and traffic hotspot scenarios. However, most existing approaches focus on optimizing communication-centric metrics such as coverage, throughput, and Call Success

Rate (CSR) while neglecting the practical energy limitations of UAV systems. The UAV-BS may continuously reposition for marginal performance gains, depleting its battery and limiting deployment lifetime. This motivates an energy-aware positioning framework that balances communication quality with movement cost while still maintaining reliable wireless service.

Chapter 2

Literature Review

2.1 Base Paper Overview

The primary reference for this work is the paper titled *"Online Positioning of a Drone-Mounted Base Station in Emergency Scenarios"* [1]. The paper proposes a real-time, data-driven positioning algorithm for UAV-BSs operating in emergency communication scenarios such as natural disasters, infrastructure failures, and sudden traffic hotspot events.

The base paper addresses the problem of dynamically positioning a UAV-BS to restore wireless coverage and improve network capacity when terrestrial infrastructure becomes unavailable or overloaded. Unlike conventional optimization approaches that require complete environmental knowledge or offline training, the proposed method operates online using real-time network measurements.

The authors analyze several control parameters including UAV horizontal position, altitude, and Cell Individual Offset (CIO). Through simulation-based sensitivity analysis, the study concludes that horizontal positioning has the greatest impact on network performance, while altitude and CIO have comparatively smaller influence on the CSR. As a result, the proposed online algorithm primarily adjusts the UAV-BS's horizontal position.

A major contribution of the base paper is the evaluation of multiple control metrics (CMs) for UAV movement decisions. Among the evaluated metrics, CM7 demonstrates the strongest correlation with CSR performance. CM7 represents the average load per user across the drone cell and the most heavily loaded neighboring cell, weighted by the number of assigned users. Using this metric, the UAV-BS continuously adjusts its position toward regions that improve network performance.

The paper demonstrates that online UAV-BS positioning significantly improves CSR compared to static deployment strategies, and that the algorithm can dynamically adapt to moving traffic hotspots without requiring prior knowledge of the environment. However, the algorithm does not account for UAV propulsion energy consumption [1].

2.2 Related Research

Several studies have explored UAV-assisted wireless communication systems, focusing on positioning optimization, coverage enhancement, and energy efficiency.

One important supporting work is *"Energy-aware Relay Positioning in Flying Networks"* [2]. This paper addresses relay positioning of Flying Communication Relay (FCR) UAVs under propulsion energy constraints. The authors highlight that UAV propulsion energy dominates communication energy consumption and demonstrate that hovering is not always the most energy-efficient operating state.

The paper proposes the **Energy-aware Relay Positioning (EREP)** algorithm, which determines UAV trajectories and movement speeds that minimize propulsion energy while maintaining communication quality. The work models propulsion power using blade profile, induced, and parasite power components. Simulation results show significant improvements in UAV endurance with only negligible degradation in throughput and delay performance. However, the proposed framework assumes a controlled network topology and relies on predefined environmental information, making it less suitable for lightweight real-time adaptive deployment [2].

Another relevant direction involves reinforcement learning-based UAV positioning approaches. Methods such as *Efficient 3D Aerial Base Station Placement Considering Users Mobility by Reinforcement Learning* [3] employ machine learning algorithms to dynamically optimize UAV placement under changing user mobility conditions. These approaches can adapt to complex environments and varying traffic distributions. However, they typically require extensive training data, high computational resources, and long convergence times, which may limit their applicability in emergency scenarios requiring rapid deployment.

2.3 Existing Challenges

Despite significant advances in UAV-assisted wireless communication, two core gaps remain unresolved.

First, most online UAV positioning algorithms optimize communication-centric metrics — throughput, coverage probability, or CSR — while neglecting the propulsion energy consumed during movement. Since UAV-BSs are battery-constrained systems, continuous repositioning can rapidly deplete onboard energy and reduce operational lifetime. Although energy-aware methods such as EREP successfully incorporate propulsion models, they generally assume known network topologies or predefined trajectories, limiting their ability to operate in fully online and dynamic environments.

Second, a clear trade-off exists between communication performance and energy efficiency. Aggressive UAV-BS movement may marginally improve CSR but substantially increases propulsion energy consumption. Designing a lightweight real-time framework that balances this trade-off remains an open problem [1, 2].

Chapter 3

Problem Statement and Research Gap

3.1 Problem Statement

In dynamic wireless scenarios such as emergency response situations and traffic hotspot events, UAV-BSs are deployed to provide temporary coverage and capacity enhancement. Existing online positioning algorithms [1] dynamically reposition the UAV-BS to maximize coverage and/or CSR. However, these approaches result in continuous UAV movement in pursuit of marginal performance improvements.

Since UAV-BSs are battery-constrained systems, frequent repositioning introduces significant propulsion energy consumption and reduces operational efficiency. A positioning mechanism is therefore needed that can balance communication performance with the energy cost associated with UAV movement.

3.2 Research Gap

Existing work has proposed adaptive UAV positioning techniques for wireless communication systems. These approaches focus primarily on optimizing network-centric metrics such as coverage probability, throughput, and CSR while assuming unlimited UAV energy availability. As a result, the movement cost of the UAV-BS is often neglected during positioning decisions.

Although some energy-aware UAV positioning approaches exist, as discussed in the previous chapter, many rely on prior environmental knowledge, offline optimization, or computationally expensive learning-based methods that are less suitable for lightweight real-time deployment.

A practical research gap therefore exists in developing a real-time, energy-aware UAV positioning framework that can dynamically adapt to changing network conditions while incorporating movement energy constraints into the positioning decision process.

3.3 Objectives

The primary objective of this work is to develop and evaluate an energy-aware online UAV positioning framework for dynamic wireless scenarios involving network failures and traffic hotspots.

The specific objectives are as follows:

- To study and analyze the existing online UAV positioning technique for emergency wireless communication systems proposed in [1].
- To evaluate the performance of different control metrics for adaptive UAV positioning across urban and rural deployment scenarios.
- To incorporate movement energy cost into the UAV positioning decision process to reduce excess UAV repositioning.
- To analyze the trade-off between UAV movement reduction and CSR performance under varying network conditions.

Chapter 4

System Model

4.1 Network Architecture

The simulated wireless communication system consists of a hexagonal cellular network containing 12 three-sectorized base station sites, resulting in a total of 36 cells. To reduce edge effects and emulate a large-scale deployment environment, the network employs wraparound boundary conditions.

Under normal operating conditions, all terrestrial base stations remain active and provide wireless connectivity to users distributed across the network area. To model an emergency communication scenario, one base station site is assumed to fail, creating a coverage hole and reduced service availability in the affected region. The failing site is selected randomly to avoid selection bias. A UAV-BS is subsequently deployed to provide temporary coverage assistance and improve network performance.

The UAV-BS functions as a mobile aerial base station operating at a fixed altitude. It dynamically repositions itself in the horizontal plane according to the positioning algorithm to adapt to evolving conditions. As demonstrated in [1], altitude and CIO optimization does not significantly affect the final CSR performance; the drone's altitude is therefore fixed at 120 m and the CIO is set to 0 dB. Both urban and rural deployment environments are considered to evaluate system behavior under different propagation and traffic characteristics.

The network additionally includes traffic hotspot regions with increased user density and communication demand. These hotspots represent realistic situations such as emergency response areas, crowded public events, or disaster-affected zones where temporary network congestion may occur.

Scenarios are represented in the format $\{DU,\ RU\}$-d_{hotspot}-ϕ_{hotspot}-ρ. For example, "DU-100-90-4" refers to a scenario with a dense urban environment, a traffic hotspot located 100 m from the failing site at an angle of 90 degrees with respect to the x-axis, and a traffic intensity four times higher than elsewhere.

4.2 Communication Model

The communication framework considers both terrestrial and air-to-ground wireless links. Communication between terrestrial base stations and user equipment (UE) follows standard cellular propagation models, while communication between the UAV-BS and users is modeled using an air-to-ground propagation model incorporating both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) conditions.

For terrestrial links, path loss is determined using standardized urban and rural propagation models [6]. For UAV communication links, the path loss model additionally considers environmental characteristics such as building density, built-up area ratio, and building height distribution. The probability of LoS communication depends on the elevation angle between the UAV-BS and the UE. Path loss calculations further incorporate shadow fading effects and minimum coupling loss to model realistic wireless channel conditions.

Users are associated with the cell providing the strongest received signal strength. Coverage is determined using Reference Signal Received Power (RSRP), and users are admitted only when sufficient network resources are available to satisfy the minimum bitrate requirement. Resource allocation is performed using a proportional fair scheduling mechanism [6] to ensure that active users maintain the required communication quality.

The system additionally incorporates admission control, handover support, and interference-aware Signal-to-Interference-plus-Noise Ratio (SINR) estimation to dynamically manage network resources under changing traffic conditions. Drone movement may alter signal strength and load distribution across cells, resulting in handovers, reassociation events, or resource reallocation during operation.

4.3 Performance Metrics

The primary performance metric used in this work is the Call Success Rate (CSR), defined as the fraction of user calls that are successfully admitted and maintained while satisfying the minimum communication requirements throughout the entire call duration. A call is considered successful only if it has coverage, is admitted to the system, is not dropped, and receives at least the minimum required bitrate for its full duration.

To evaluate the efficiency of the proposed positioning approach, UAV movement behavior is also analyzed. The total number of movement steps performed by the UAV-BS is used as an approximate indicator of propulsion energy consumption. A reduction in excess movement corresponds to improved energy efficiency and enhanced operational stability.

The proposed energy-aware positioning framework is compared against the baseline online positioning method in terms of both CSR and movement efficiency.

4.4 Modeling Assumptions

Several assumptions are adopted to simplify the simulation environment while preserving realistic system behavior. The UAV-BS operates at a fixed altitude throughout the simulation and moves only in the horizontal plane using discrete movement steps determined by the positioning algorithm. User equipment remains static during active communication sessions.

A simplified discrete energy model is employed in which UAV movement consumes fixed energy values per simulation step. The model focuses on propulsion-related energy consumption and does not explicitly consider aerodynamic effects, acceleration dynamics, or communication hardware power consumption in detail, as addressed in [2].

The simulations are conducted under both urban and rural deployment environments using predefined propagation parameters, hotspot configurations, and traffic generation models. Multiple simulation runs are averaged to reduce randomness and improve the consistency of the observed results.

Chapter 5

Proposed Energy-Aware Improvement

5.1 Baseline Positioning Method

The baseline is the online drone positioning algorithm proposed in [1]. The algorithm dynamically adjusts the horizontal position of the UAV-BS using real-time network measurements to improve CSR during emergency scenarios.

The original framework operates by iteratively moving the UAV-BS in the horizontal plane and evaluating the effect of each movement using a control metric (CM). The UAV-BS continuously adjusts its position in the direction that improves the selected CM, allowing the drone to adapt to changing user distributions and traffic hotspot conditions without requiring prior environmental knowledge or offline training.

Among the evaluated control metrics, CM7 was identified as the most effective for approximating the CSR. CM7 represents the average load per user across the drone cell and the most heavily loaded neighboring cell, weighted by the number of users assigned to each cell. In this context, *cell load* is defined as the fraction of total downlink resources that a cell must allocate to provide all its assigned users their minimum required bitrate. Weighting by user count ensures that cells with more users have proportionally greater influence on the combined metric, enabling more balanced positioning decisions.

Although the baseline approach achieves strong CSR performance, it continuously repositions the UAV-BS in pursuit of small performance gains and does not consider the propulsion energy consumed during movement.

5.2 Energy Consumption Model

To incorporate energy awareness into the positioning process, a simplified discrete energy model is introduced. The model approximates propulsion-related energy consumption using fixed energy costs associated with UAV movement and hovering operations.

At each simulation step, the UAV-BS consumes a fixed amount of energy denoted as E. Although this does not explicitly model velocity variation, acceleration, or communication hardware power consumption, it effectively captures the relative energy cost associated with excess UAV movement. Since positioning is restricted to the horizontal plane, gravitational and variable hovering costs remain constant and need not be optimized.

5.3 Proposed Algorithm

The proposed approach extends the baseline online positioning framework by introducing energy-aware decision making into the movement process. Instead of moving solely to maximize CSR, the UAV-BS evaluates whether the expected performance improvement justifies the corresponding movement energy cost.

The algorithm continues to use the CM7 control metric as the primary indicator of network performance, which is validated as the best CSR proxy in Chapter 7. At each decision step, the UAV-BS measures the change in the control metric after movement and computes the corresponding performance gain.

The gain is defined as:

$$\text{Gain} = CM7_{old} - CM7_{new}$$

A movement is accepted only when:

$$\text{Gain} > \lambda \times \text{Energy} + \epsilon$$

where:

- λ is the trade-off parameter controlling the balance between performance and energy efficiency,
- ϵ is a small threshold used to prevent excess movement caused by minor fluctuations or measurement noise.

If the observed improvement is insufficient, the UAV-BS avoids unnecessary repositioning and either remains stationary or explores an alternative direction. This allows the drone to stabilize near efficient operating positions instead of continuously chasing marginal performance improvements.

5.4 Algorithm Workflow

Figure 5.1 illustrates the workflow of the proposed algorithm [1].

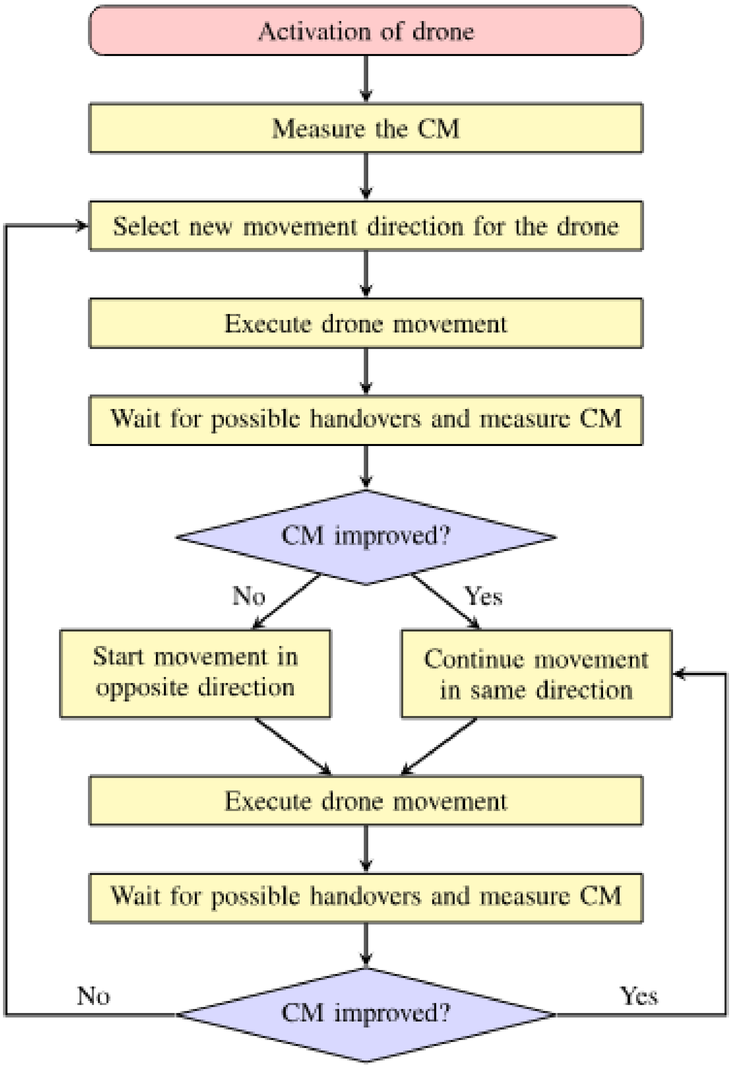

Figure 5.1: Workflow of the proposed energy-aware UAV positioning algorithm

Chapter 6

Simulation Setup

6.1 Simulation Environment

The proposed UAV positioning framework is evaluated using the network architecture described in Chapter 4. The simulation models an emergency communication scenario in which one terrestrial base station site fails, creating a coverage hole within the network. Following the failure event, a UAV-BS is deployed to provide temporary coverage support and dynamically reposition itself according to the selected positioning algorithm.

The simulation process consists of multiple operational phases. Initially, the network operates under normal conditions to establish stable traffic behavior. A base station failure is then introduced, after which the UAV-BS is activated and begins dynamic positioning using either the baseline or the proposed energy-aware positioning algorithm.

6.2 Simulation Parameters

The simulation parameters are drawn from [1] with additions for the energy-aware extension. The UAV-BS operates at a fixed altitude and performs discrete horizontal movement steps during online positioning. Real-time control decisions are evaluated periodically using the CM7 control metric.

Table 6.1 summarizes the key simulation parameters used throughout the evaluation.

Table 6.1: Simulation Parameters

Parameter	**Urban**	**Rural**	**Unit**
ISD	500	3500	m
h_{UE}	1.5	1.5	m
Carrier Frequency	3.5	3.5	GHz
Bandwidth per Cell	5	5	MHz
α	0.5	0.05	–
β	300	300	buildings/km^2
γ	20	4	–
λ	1.3	0.018	arrivals/s/km^2
τ	120	120	s
κ	0.25	0.25	–
R	0.384	0.384	Mb/s
G_r	15.4	15.4	dBi
HPBW_h	72	72	$^\circ$
HPBW_v	5.6	5.6	$^\circ$
FBR_h	24	24	dB
SLL_v	-11	-11	dB
θ_{etilt}	7	4	$^\circ$
G_d	8	8	dBi
HPBW_d	65	65	$^\circ$
$A_{\max}$	30	30	dB
η_{LoS}	1.7	0.1	dB
η_{NLoS}	24	22	dB
d_{decorr}	45	80	m
σ_{sh}	5	6	dB
ω	0.5	0.5	–
UAV Altitude	120	120	m
Drone Step Size	1	1	m
Drone Speed	24	24	km/h
Measurement Period	200	200	ms
Control Interval	400	400	ms
Coverage Threshold	-120	-120	dBm
Hotspot Radius	100	100	m
Simulation Runs	25	25	–
Energy Cost Parameter (E)	0.01	0.01	–
Threshold Parameter (ϵ)	0.001	0.001	–

The urban deployment scenarios use hotspot intensity multipliers of 2, 4, and 8, while the rural scenarios use multipliers of 10, 25, and 50 to model varying traffic congestion conditions [1]. Multiple simulation runs are averaged to reduce randomness and improve result consistency.

6.3 Evaluation Criteria

The performance of the proposed energy-aware positioning framework is evaluated by comparing it against the baseline online UAV positioning algorithm under identical network conditions.

The primary evaluation metric used in this work is the Call Success Rate (CSR), which represents the fraction of user calls successfully admitted and maintained while satisfying the minimum communication requirements. CSR is used to evaluate the overall communication effectiveness of the positioning strategy under different deployment scenarios.

In addition, UAV movement behavior is analyzed to evaluate operational efficiency. The total number of movement steps performed by the UAV-BS is used as an approximate indicator of propulsion-related energy consumption. Lower movement counts correspond to reduced energy expenditure and improved deployment stability.

Performance comparisons are conducted across multiple urban and rural deployment scenarios with varying hotspot intensities and failure conditions. The evaluation includes comparisons between:

- No-drone deployment scenarios,
- Static UAV-BS deployment,
- Baseline online positioning methods using different control metrics,
- The proposed energy-aware CM7 positioning framework.

The analysis further considers convergence behavior and the trade-off between CSR and movement efficiency across different network environments.

Chapter 7

Performance Evaluation, Results, and Discussion

This chapter presents the experimental evaluation of the proposed energy-aware UAV positioning framework. The analysis is organized into three primary stages. First, the effectiveness of the different control metrics for online UAV positioning is evaluated across urban and rural deployment scenarios, verifying the results reported in [1]. Next, the temporal convergence behavior of the positioning framework is analyzed to study operational stability after UAV deployment, and the trends are matched against the experimental results in [1]. Finally, the energy-aware positioning strategy is evaluated in terms of the trade-off between CSR and UAV movement efficiency.

7.1 Control Metric Evaluation

7.1.1 Control Metrics Defined

The seven control metrics evaluated in this work are drawn from [1] and are based on the concept of *cell load*, defined as the fraction of total downlink resources a cell must allocate to provide all its assigned users their minimum required bitrate. The metrics differ in which cell(s) are observed and whether per-user load or raw load is considered:

- **CM1** — Load of the drone cell.
- **CM2** — Load of the most heavily loaded regular (terrestrial) cell adjacent to the failed site.
- **CM3** — Sum of the drone cell load and the most heavily loaded regular cell load.
- **CM4** — Average load *per user* of the drone cell.
- **CM5** — Average load per user of the most heavily loaded regular cell.

- **CM6** — Average load per user across the drone cell and the most heavily loaded regular cell, with each cell weighted equally.

- **CM7** — Average load per user across the drone cell and the most heavily loaded regular cell, with each cell weighted by its number of assigned users.

In all cases the UAV-BS moves in the direction that minimizes the selected CM, under the assumption that reducing cell load (or load per user) corresponds to reduced blocking and improved CSR. CM7 provides the most accurate weighting because cells carrying more users contribute proportionally more to the combined metric, enabling finer load balancing.

7.1.2 Results

This section compares the seven evaluated control metrics against two reference deployment configurations:

- No-drone deployment,

- Static drone deployment.

The comparison is performed across multiple urban (DU) and rural (RU) deployment scenarios with varying hotspot intensities and failure conditions.

Figures 7.1 and 7.2 illustrate the normalized performance of the evaluated positioning methods across urban and rural deployment scenarios, respectively.

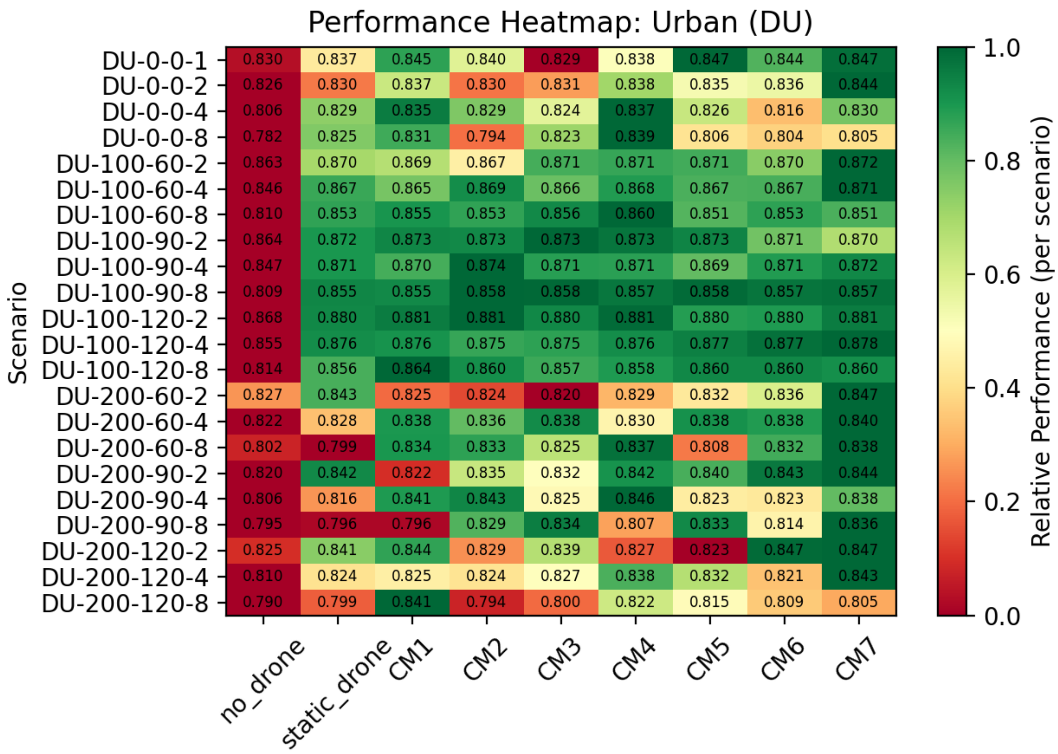

Figure 7.1: Performance comparison of control metrics across urban deployment scenarios

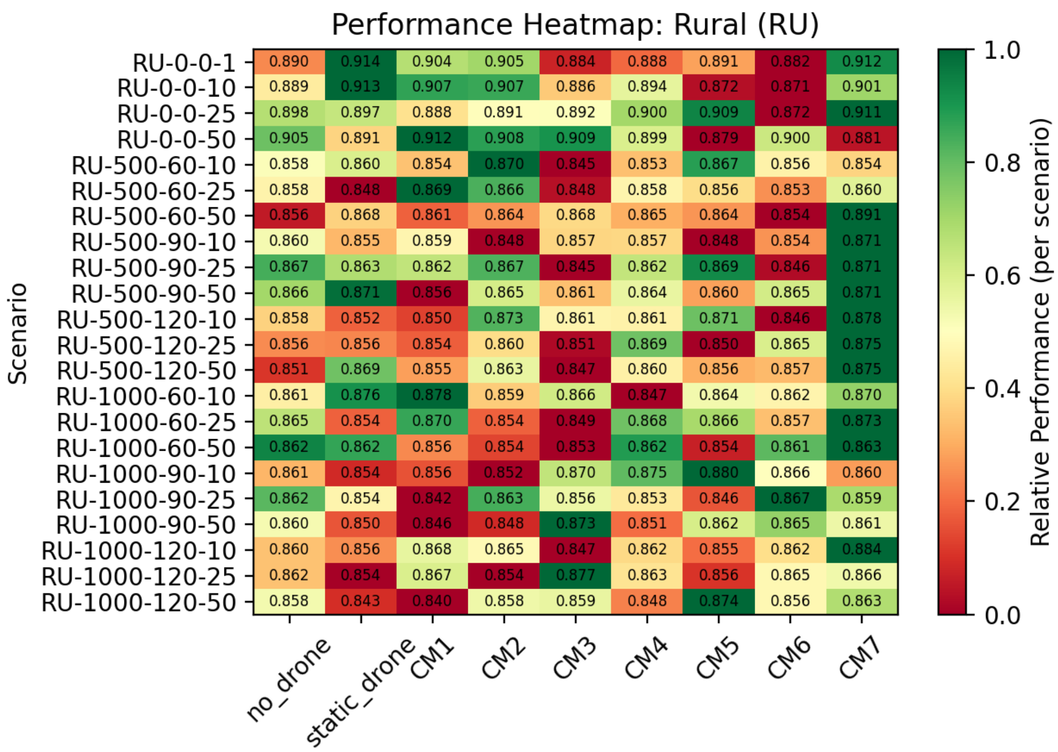

Figure 7.2: Performance comparison of control metrics across rural deployment scenarios

The results demonstrate that dynamic UAV positioning consistently improves network performance compared to both the no-drone and static-drone configurations. The performance improvement is especially visible in scenarios with severe traffic hotspots and larger coverage disruptions, where adaptive UAV repositioning allows the network to better respond to changing user demand.

Among the evaluated control metrics, CM7 demonstrates the most stable and consistently strong performance across urban deployment environments. Metrics such as CM1 and CM2 perform adequately under moderate traffic conditions but exhibit larger fluctuations under highly congested scenarios. Rural scenarios exhibit higher variance across metrics, as the larger hotspot area amplifies the coverage problem. CM7 achieves a more balanced distribution between drone load and neighboring cell load, enabling improved CSR under varying conditions.

Figure 7.3 further compares the CSR performance of the no-drone, static-drone, and CM7-based positioning approaches for individual deployment scenarios.

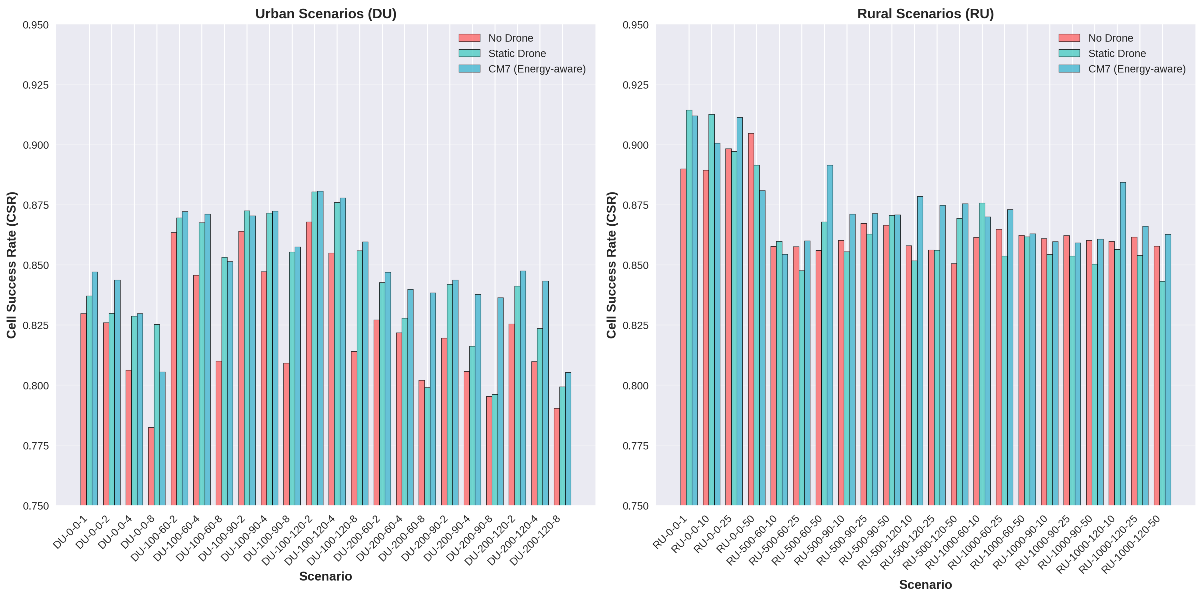

Figure 7.3: CSR comparison across urban and rural deployment scenarios

The grouped comparison confirms that the CM7-based positioning framework consistently achieves higher CSR values than the no-drone and static-drone configurations across most scenarios. The improvement is particularly noticeable in urban hotspot environments where adaptive repositioning enables stronger load balancing and more efficient user coverage.

The results additionally indicate that rural deployment scenarios generally achieve higher absolute CSR values than urban environments. This is attributable to lower interference levels and wider spatial user distribution in rural networks. In contrast, urban scenarios exhibit stronger performance variation due to higher traffic density and increased inter-cell interference.

Overall, the control metric evaluation establishes CM7 as the most effective and stable metric for online UAV positioning across varying deployment conditions, consistent with the findings in [1]. Consequently, CM7 is selected as the primary control metric for the proposed energy-aware positioning framework.

7.2 Temporal Convergence Behavior

The second stage of evaluation analyzes the convergence characteristics of the UAV positioning framework by observing the variation of CSR over time following UAV deployment. This analysis is performed in order to study the operational stability of the positioning algorithm under representative urban and rural deployment scenarios.

Figure 7.4 illustrates the convergence behavior of multiple deployment scenarios aligned at the time of UAV activation.

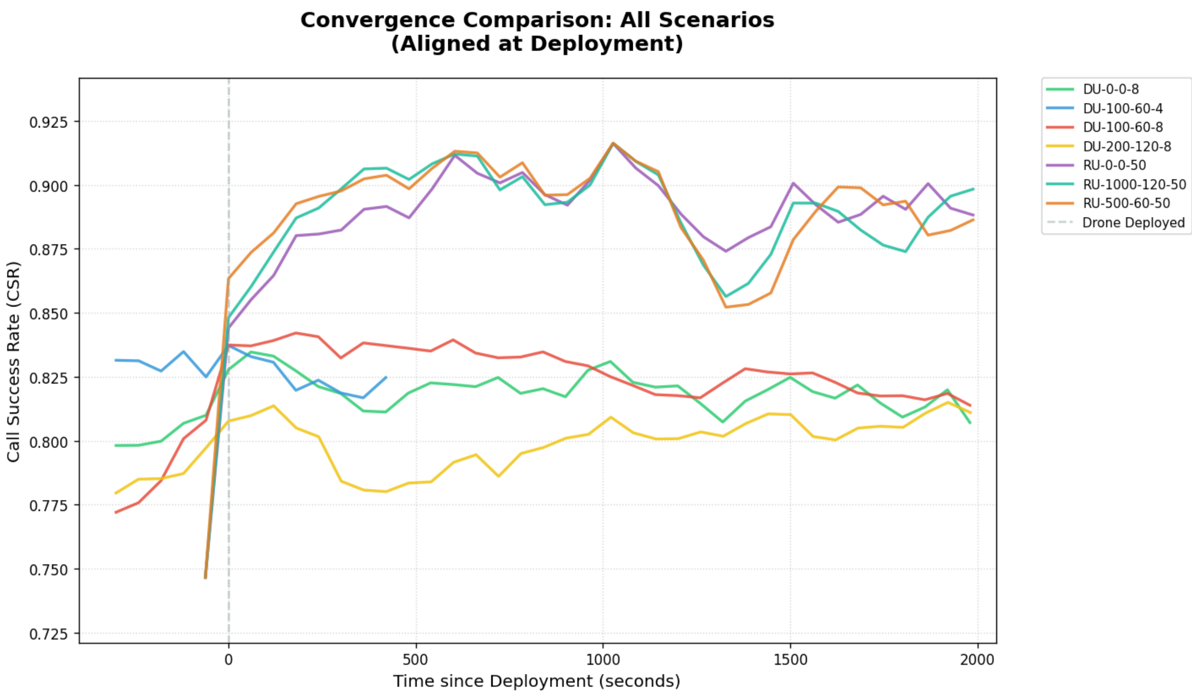

Figure 7.4: Convergence behavior of the UAV positioning framework across multiple deployment scenarios

The results show that UAV deployment immediately improves communication performance following the occurrence of a site failure. In several scenarios, CSR increases rapidly after deployment and gradually almost stabilizes near a steady operating point as the UAV adapts to changing network conditions.

The convergence curves further demonstrate that the positioning framework maintains relatively stable CSR values after the initial adaptation phase. Although temporary fluctuations occur due to changing hotspot intensity and network load distribution, the algorithm successfully avoids large oscillatory behavior and converges toward stable operating conditions.

Urban deployment scenarios generally exhibit greater CSR variability due to stronger interference effects and higher traffic concentration. Rural scenarios, on the other hand, display comparatively smoother convergence behavior because of lower interference and more stable user distribution patterns.

The convergence analysis therefore indicates that the proposed positioning framework is capable of adapting dynamically to changing wireless conditions while maintaining stable long-term CSR, consistent with the behavior reported in [1].

7.3 Energy-Aware Positioning Trade-off

The final stage of evaluation analyzes the effectiveness of the proposed energy-aware positioning strategy in reducing unnecessary UAV movement while preserving communication performance.

Figure 7.5 presents the relationship between the number of movement steps saved relative to the legacy positioning algorithm and the corresponding CSR difference between the energy-aware and legacy approaches.

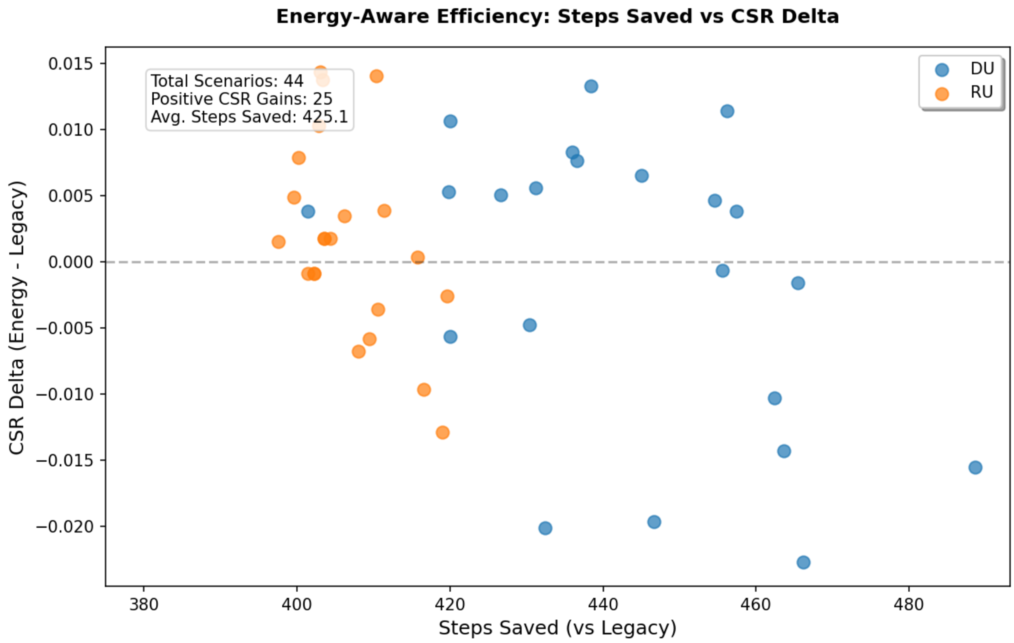

Figure 7.5: Movement efficiency versus CSR difference for the proposed energy-aware positioning framework

The results indicate that the proposed energy-aware framework achieves substantial reductions in UAV movement across nearly all evaluated scenarios. On average, more than 400 movement steps are eliminated relative to the legacy positioning approach, significantly reducing propulsion-related energy expenditure.

At the same time, the observed CSR variation remains relatively small across most deployment scenarios. Several cases even demonstrate slight CSR improvements despite the reduction in movement, indicating that continuous repositioning is not always necessary for maintaining effective communication performance.

The analysis additionally reveals a clear trade-off between communication performance and movement efficiency. While a limited number of scenarios experience minor CSR degradation, the reduction in UAV movement remains substantially larger in comparison, resulting in improved overall operational efficiency.

Rural deployment scenarios generally achieve movement reduction with smaller CSR degradation compared to highly dynamic urban environments. This behavior occurs because rural deployments experience lower interference variability and more stable user distributions, allowing the UAV to remain near efficient operating positions for longer durations.

Overall, the results demonstrate that the proposed energy-aware positioning framework successfully balances communication performance and operational efficiency by reducing unnecessary UAV movement while maintaining near-equivalent CSR performance.

7.4 Overall Discussion

The combined evaluation results demonstrate that adaptive UAV positioning significantly improves communication performance compared to conventional no-drone and static-drone deployment strategies. Among the evaluated positioning metrics, CM7 consistently provides the strongest and most stable performance across varying urban and rural deployment conditions.

The convergence analysis further confirms that the proposed framework maintains stable operational behavior after deployment and successfully adapts to changing network conditions without large performance oscillations.

Most importantly, the proposed energy-aware extension substantially reduces unnecessary UAV movement while preserving near-equivalent CSR performance. This indicates that continuously repositioning the UAV for marginal performance gains is not always operationally beneficial, particularly in practical battery-constrained deployment scenarios.

The results therefore validate the effectiveness of incorporating energy-awareness into online UAV positioning decisions and demonstrate the feasibility of balancing communication performance with operational efficiency in dynamic wireless communication systems.

Chapter 8

Limitations

Although the proposed energy-aware UAV positioning framework demonstrates promising improvements in operational efficiency, the current implementation includes several simplifying assumptions and constraints.

The adopted energy model is intentionally oversimplified and primarily focuses on propulsion-related movement cost. The framework does not explicitly model detailed UAV battery dynamics, aerodynamic effects, acceleration behavior, wind conditions, or communication hardware power consumption. As a result, the estimated energy expenditure serves only as an approximate representation of practical UAV energy usage.

The study considers only a single UAV-mounted base station operating within the network. Multi-UAV coordination, collision avoidance, cooperative load balancing, and inter-UAV communication are not included in the current framework.

Furthermore, the evaluation is performed within a simulation environment using predefined traffic models, hotspot configurations, and propagation assumptions. Real-world deployment conditions may introduce additional challenges such as unpredictable mobility patterns, hardware limitations, environmental obstacles, and communication delays.

Despite these limitations, the proposed framework provides a practical and computationally lightweight foundation for energy-aware online UAV positioning in dynamic wireless communication systems.

Chapter 9

Future Scope

Several extensions can be explored in future work to further improve the effectiveness and practicality of the proposed UAV positioning framework.

- Extension of the framework to support coordinated multi-UAV deployment and cooperative positioning strategies.
- Integration of realistic UAV battery and propulsion models incorporating aerodynamic effects and environmental conditions.
- Development of adaptive altitude optimization methods for improved energy usage.
- Evaluation under dynamic user mobility scenarios and time-varying traffic distributions.
- Validation of the proposed framework using hardware-based UAV platforms and real-world deployment environments.

Chapter 10

Conclusion

This work has demonstrated that integrating energy-awareness into online UAV-BS positioning leads to substantial reductions in unnecessary movement with minimal impact on CSR. The proposed framework — built on the CM7 control metric and a lightweight gain-threshold criterion — consistently outperforms the unmodified baseline in terms of operational efficiency across urban and rural deployment scenarios, while maintaining near-equivalent service quality.

The key finding is that continuous UAV repositioning for marginal performance gains is operationally unnecessary: a significant fraction of movement steps can be eliminated without degrading the CSR, extending battery life and improving practical deployment viability. CM7 was confirmed as the most reliable proxy for CSR among the seven evaluated metrics, and the proposed energy-aware extension preserves this advantage while adding movement economy.

These results provide a practical, computationally lightweight foundation for energy-aware aerial network management and motivate future work on multi-UAV coordination, realistic propulsion modeling, and hardware validation.

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